In [1]:
```
%matplotlib inline
```

Reinforcement Learning (DQN) tutorial

Task

The agent has to decide between two actions - moving the cart left or right - so that the pole attached to it stays upright. You can find an official leaderboard with various algorithms and visualizations at the `Gym website <https://gym.openai.com/envs/CartPole-v1>` .(Currently `gymnasium`).

As the agent observes the current state of the environment and chooses an action, the environment *transitions* to a new state, and also returns a reward that indicates the consequences of the action. In this task, the environment terminates if the pole falls over too far.

The CartPole task is designed so that the inputs to the agent are 4 real values representing the environment state (position, velocity, etc.). However, neural networks can solve the task purely by looking at the scene, so we'll use a patch of the screen centered on the cart as an input. Because of this, our results aren't directly comparable to the ones from the official leaderboard - our task is much harder. Unfortunately this does slow down the training, because we have to render all the frames.

Strictly speaking, we will present the state as the difference between the current screen patch and the previous one. This will allow the agent to take the velocity of the pole into account from one image.

Packages

First, let's import needed packages. Firstly, we need `gymnasium` for the environment (Install using `pip install gymnasium`). We'll also use the following from PyTorch:

- neural networks (`torch.nn`)
- optimization (`torch.optim`)
- automatic differentiation (`torch.autograd`)
- utilities for vision tasks (`torchvision` - `a separate package <https://github.com/pytorch/vision>`).

In [2]:
```
import gymnasium as gym
import math
import random
import numpy as np
import matplotlib
import matplotlib.pyplot as plt
from collections import namedtuple
from itertools import count
from PIL import Image

import torch
```

```
import torch.nn as nn
import torch.optim as optim
import torch.nn.functional as F
import torchvision.transforms as T

env = gym.make('CartPole-v1',render_mode='rgb_array').unwrapped

# set up matplotlib
is_ipython = 'inline' in matplotlib.get_backend()
if is_ipython:
    from IPython import display

plt.ion()

# if gpu is to be used
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
```

Replay Memory

We'll be using experience replay memory for training our DQN. It stores the transitions that the agent observes, allowing us to reuse this data later. By sampling from it randomly, the transitions that build up a batch are decorrelated. It has been shown that this greatly stabilizes and improves the DQN training procedure.

For this, we're going to need two classses:

- `Transition` - a named tuple representing a single transition in our environment
- `ReplayMemory` - a cyclic buffer of bounded size that holds the transitions observed recently. It also implements a `.sample()` method for selecting a random batch of transitions for training.

In [3]:
```
Transition = namedtuple('Transition',
                        ('state', 'action', 'next_state', 'reward'))

class ReplayMemory(object):

    def __init__(self, capacity):
        self.capacity = capacity
        self.memory = []
        self.position = 0

    def push(self, *args):
        """Saves a transition."""
        if len(self.memory) < self.capacity:
            self.memory.append(None)
        self.memory[self.position] = Transition(*args)
        self.position = (self.position + 1) % self.capacity

    def sample(self, batch_size):
        return random.sample(self.memory, batch_size)

    def __len__(self):
        return len(self.memory)
```

Now, let's define our model. But first, let quickly recap what a DQN is.

DQN algorithm

Our environment is deterministic, so all equations presented here are also formulated deterministically for the sake of simplicity. In the reinforcement learning literature, they would also contain expectations over stochastic transitions in the environment.

Our aim will be to train a policy that tries to maximize the discounted, cumulative reward $R_{t_0} = \sum_{t=t_0}^{\infty} \gamma^{t-t_0} r_t$, where R_{t_0} is also known as the *return*. The discount, γ, should be a constant between 0 and 1 that ensures the sum converges. It makes rewards from the uncertain far future less important for our agent than the ones in the near future that it can be fairly confident about.

The main idea behind Q-learning is that if we had a function $Q^* : State \times Action \to \mathbb{R}$, that could tell us what our return would be, if we were to take an action in a given state, then we could easily construct a policy that maximizes our rewards:

$$\pi^*(s) = \underset{a}{\operatorname{argmax}}\ Q^*(s, a) \tag{1}$$

However, we don't know everything about the world, so we don't have access to Q^*. But, since neural networks are universal function approximators, we can simply create one and train it to resemble Q^*.

For our training update rule, we'll use a fact that every Q function for some policy obeys the Bellman equation:

$$Q^\pi(s, a) = r + \gamma Q^\pi(s', \pi(s')) \tag{2}$$

The difference between the two sides of the equality is known as the temporal difference error, δ:

$$\delta = Q(s, a) - (r + \gamma \max_a Q(s', a)) \tag{3}$$

To minimise this error, we will use the `Huber loss <https://en.wikipedia.org/wiki/Huber_loss>`. The Huber loss acts like the mean squared error when the error is small, but like the mean absolute error when the error is large - this makes it more robust to outliers when the estimates of Q are very noisy. We calculate this over a batch of transitions, B, sampled from the replay memory:

$$\mathcal{L} = \frac{1}{|B|} \sum_{(s,a,s',r)\ \in\ B} \mathcal{L}(\delta) \tag{4}$$

$$\text{where} \quad \mathcal{L}(\delta) = \begin{cases} \frac{1}{2}\delta^2 & \text{for } |\delta| \le 1, \\ |\delta| - \frac{1}{2} & \text{otherwise.} \end{cases} \tag{5}$$

Q-network

Our model will be a convolutional neural network that takes in the difference between the current and previous screen patches. It has two outputs, representing $Q(s, \text{left})$ and

$Q(s, \text{right})$ (where s is the input to the network). In effect, the network is trying to predict the *quality* of taking each action given the current input.

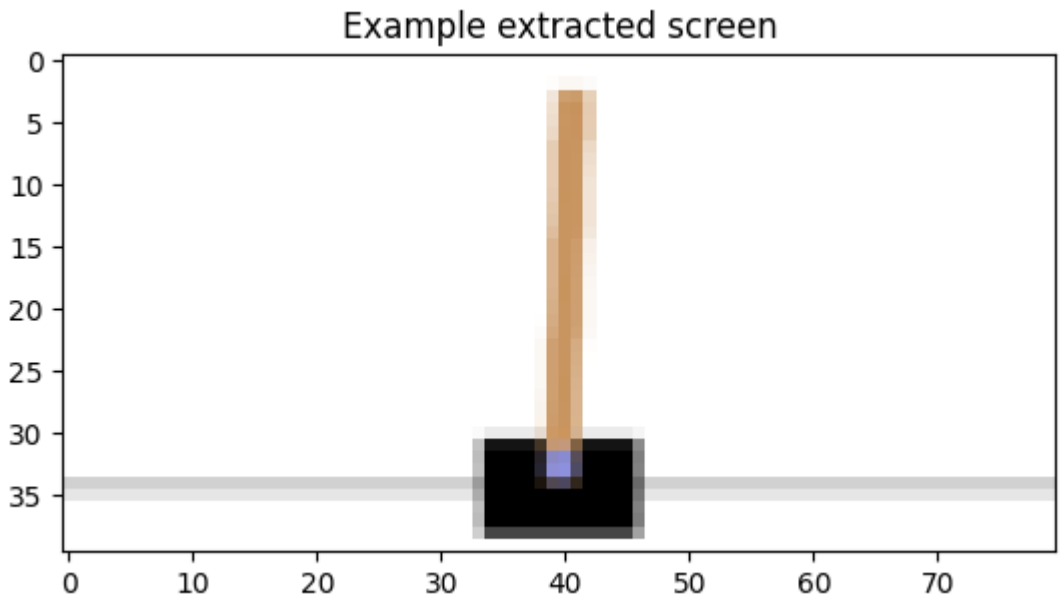
In [4]:
```
class DQN(nn.Module):

    def __init__(self):
        super(DQN, self).__init__()
        self.conv1 = nn.Conv2d(3, 16, kernel_size=5, stride=2)
        self.bn1 = nn.BatchNorm2d(16)
        self.conv2 = nn.Conv2d(16, 32, kernel_size=5, stride=2)
        self.bn2 = nn.BatchNorm2d(32)
        self.conv3 = nn.Conv2d(32, 32, kernel_size=5, stride=2)
        self.bn3 = nn.BatchNorm2d(32)
        self.head = nn.Linear(448, 2)

    def forward(self, x):
        x = F.relu(self.bn1(self.conv1(x)))
        x = F.relu(self.bn2(self.conv2(x)))
        x = F.relu(self.bn3(self.conv3(x)))
        return self.head(x.view(x.size(0), -1))
```

Input extraction

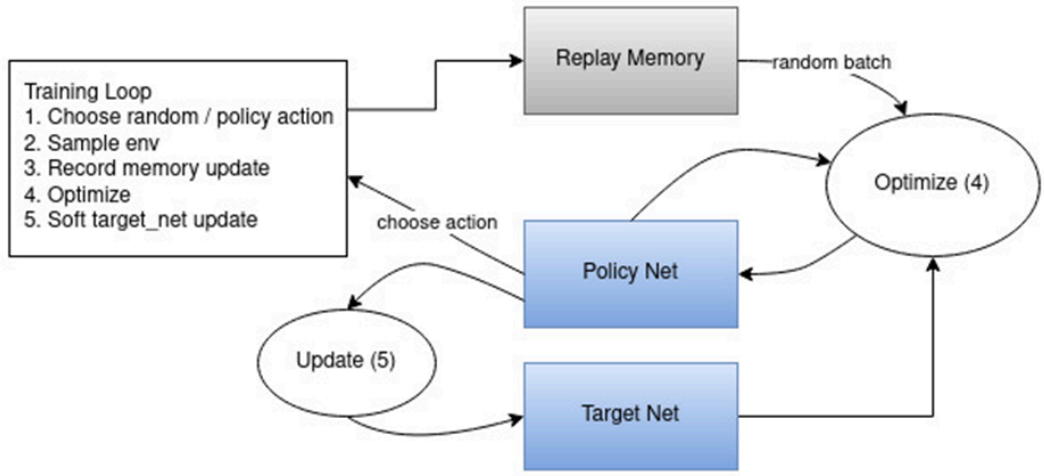
The code below are utilities for extracting and processing rendered images from the environment. It uses the `torchvision` package, which makes it easy to compose image transforms. Once you run the cell it will display an example patch that it extracted.

In [5]:
```
resize = T.Compose([T.ToPILImage(),
                    T.Resize(40, interpolation=Image.BICUBIC),
                    T.ToTensor()])

# This is based on the code from gym.
screen_width = 600

def get_cart_location():
    world_width = env.x_threshold * 2
    scale = screen_width / world_width
    return int(env.state[0] * scale + screen_width / 2.0)  # MIDDLE OF CART


def get_screen():
    screen = env.render().transpose(
        (2, 0, 1))  # transpose into torch order (CHW)
    # Strip off the top and bottom of the screen
    screen = screen[:, 160:320]
    view_width = 320
    cart_location = get_cart_location()
    if cart_location < view_width // 2:
        slice_range = slice(view_width)
    elif cart_location > (screen_width - view_width // 2):
        slice_range = slice(-view_width, None)
    else:
        slice_range = slice(cart_location - view_width // 2,
                            cart_location + view_width // 2)
    # Strip off the edges, so that we have a square image centered on a cart
    screen = screen[:, :, slice_range]
    # Convert to float, rescare, convert to torch tensor
```

```
    # (this doesn't require a copy)
    screen = np.ascontiguousarray(screen, dtype=np.float32) / 255
    screen = torch.from_numpy(screen)
    # Resize, and add a batch dimension (BCHW)
    return resize(screen).unsqueeze(0).to(device)


env.reset()
plt.figure()
plt.imshow(get_screen().cpu().squeeze(0).permute(1, 2, 0).numpy(),
           interpolation='none')
plt.title('Example extracted screen')
plt.show()
```



Training



Hyperparameters and utilities

This cell instantiates our model and its optimizer, and defines some utilities:

- `select_action` - will select an action accordingly to an epsilon greedy policy. Simply put, we'll sometimes use our model for choosing the action, and sometimes we'll just sample one uniformly. The probability of choosing a random action will start at `EPS_START` and will decay exponentially towards `EPS_END`. `EPS_DECAY` controls the rate of the decay.
- `plot_durations` - a helper for plotting the durations of episodes, along with an average over the last 100 episodes (the measure used in the official evaluations). The plot will be underneath the cell containing the main training loop, and will update after every episode.

In [6]:
```
BATCH_SIZE = 128
GAMMA = 0.999
EPS_START = 0.9
EPS_END = 0.05
EPS_DECAY = 200
TARGET_UPDATE = 10

policy_net = DQN().to(device)
target_net = DQN().to(device)
target_net.load_state_dict(policy_net.state_dict())
target_net.eval()

optimizer = optim.RMSprop(policy_net.parameters())
memory = ReplayMemory(10000)

steps_done = 0

def select_action(state):
    global steps_done
    sample = random.random()
    eps_threshold = EPS_END + (EPS_START - EPS_END) * \
        math.exp(-1. * steps_done / EPS_DECAY)
    steps_done += 1
    if sample > eps_threshold:
        with torch.no_grad():
            return policy_net(state).max(1)[1].view(1, 1)
    else:
        return torch.tensor([[random.randrange(2)]], device=device, dtype=torch.

episode_durations = []

def plot_durations():
    if is_ipython:
        display.clear_output(wait=True)
        # display.display(plt.gcf())
    plt.figure(2)
    # plt.clf()
    durations_t = torch.tensor(episode_durations, dtype=torch.float)
    plt.title('Training...')
    plt.xlabel('Episode')
    plt.ylabel('Duration')
    plt.plot(durations_t.numpy())
    # Take 100 episode averages and plot them too
    if len(durations_t) >= 100:
        means = durations_t.unfold(0, 100, 1).mean(1).view(-1)
        means = torch.cat((torch.zeros(99), means))
        plt.plot(means.numpy())
```

```
        plt.show()

    plt.pause(0.001)  # pause a bit so that plots are updated
    if is_ipython:
        # display.clear_output(wait=True)
        # display.display(plt.gcf())
        pass
```

Training loop

Finally, the code for training our model.

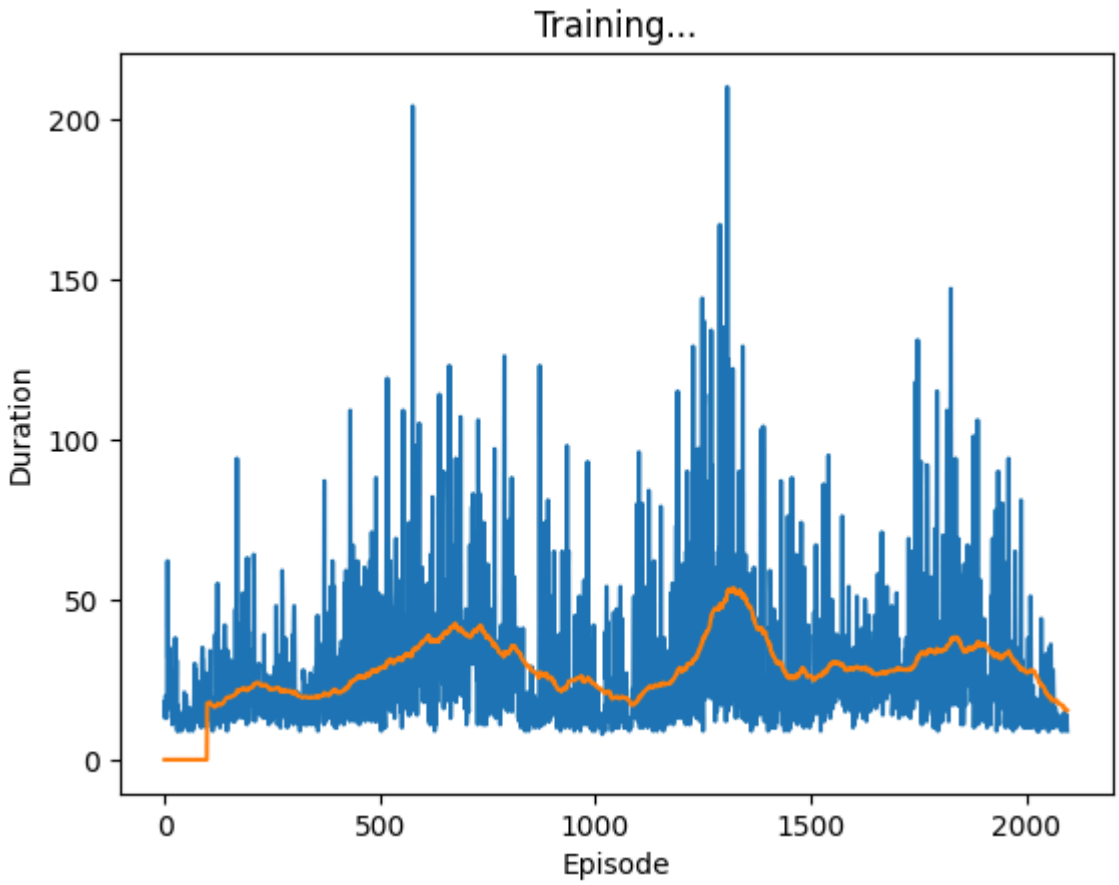
Here, you can find an `optimize_model` function that performs a single step of the optimization. It first samples a batch, concatenates all the tensors into a single one, computes $Q(s_t, a_t)$ and $V(s_{t+1}) = \max_a Q(s_{t+1}, a)$, and combines them into our loss. By defition we set $V(s) = 0$ if s is a terminal state. We also use a target network to compute $V(s_{t+1})$ for added stability. The target network has its weights kept frozen most of the time, but is updated with the policy network's weights every so often. This is usually a set number of steps but we shall use episodes for simplicity.

In [7]:
```
def optimize_model():
    if len(memory) < BATCH_SIZE:
        return
    transitions = memory.sample(BATCH_SIZE)
    # Transpose the batch (see http://stackoverflow.com/a/19343/3343043 for
    # detailed explanation).
    batch = Transition(*zip(*transitions))

    # Compute a mask of non-final states and concatenate the batch elements
    non_final_mask = torch.tensor(tuple(map(lambda s: s is not None,
                                            batch.next_state)), device=device, dty
    non_final_next_states = torch.cat([s for s in batch.next_state
                                                 if s is not None])
    state_batch = torch.cat(batch.state)
    action_batch = torch.cat(batch.action)
    reward_batch = torch.cat(batch.reward)

    # Compute Q(s_t, a) - the model computes Q(s_t), then we select the
    # columns of actions taken
    state_action_values = policy_net(state_batch).gather(1, action_batch)

    # Compute V(s_{t+1}) for all next states.
    next_state_values = torch.zeros(BATCH_SIZE, device=device)
    next_state_values[non_final_mask] = target_net(non_final_next_states).max(1)
    # Compute the expected Q values
    expected_state_action_values = (next_state_values * GAMMA) + reward_batch

    # Compute Huber loss
    loss = F.smooth_l1_loss(state_action_values, expected_state_action_values.un

    # Optimize the model
    optimizer.zero_grad()
    loss.backward()
    for param in policy_net.parameters():
        param.grad.data.clamp_(-1, 1)
    optimizer.step()
```

Below, you can find the main training loop. At the beginning we reset the environment and initialize the `state` Tensor. Then, we sample an action, execute it, observe the next screen and the reward (always 1), and optimize our model once. When the episode ends (our model fails), we restart the loop.

Below, `num_episodes` could be set small for initial debugging (i.e 50). For actual training you should run lot more epsiodes.

In [ ]:
```
num_episodes = 5000 # 50

for i_episode in range(num_episodes):
    # Initialize the environment and state
    env.reset()
    last_screen = get_screen()
    current_screen = get_screen()
    state = current_screen - last_screen
    for t in count():
        # Select and perform an action
        action = select_action(state)
        _, reward, terminated, truncated, _ = env.step(action.item())
        done = terminated or truncated
        reward = torch.tensor([reward], device=device)

        # Observe new state
        last_screen = current_screen
        current_screen = get_screen()
        if not done:
            next_state = current_screen - last_screen
        else:
            next_state = None

        # Store the transition in memory
        memory.push(state, action, next_state, reward)

        # Move to the next state
        state = next_state

        # Perform one step of the optimization (on the target network)
        optimize_model()
        if done:
            episode_durations.append(t + 1)
            plot_durations()
            break
    # Update the target network
    if i_episode % TARGET_UPDATE == 0:
        target_net.load_state_dict(policy_net.state_dict())

print('Complete')
env.render()
env.close()
plt.ioff()
plt.show()
```


Training...
Duration
Episode
0
50
100
150
200
0
500
1000
1500
2000